MATHEMATICS PROJECT WORK RECORD :: FA-4

Topic: Factorisation

1. PRELIMINARY INFORMATION

Name of the Student:	.....
Class & Section:	.....
Roll Number:	.....
Subject:	**Mathematics**
Topic:	**Factorisation**

2. AIM

To explore, understand, and apply various methods of factorising algebraic expressions and represent quadratic factorisation geometrically using an area model.

3. OBJECTIVES

- Identify common monomial factors from given algebraic expressions.
- Factorise polynomials by regrouping terms and applying standard algebraic identities.
- Factorise quadratic trinomials by splitting the middle term and verify through geometric representation.

4. MATERIALS REQUIRED

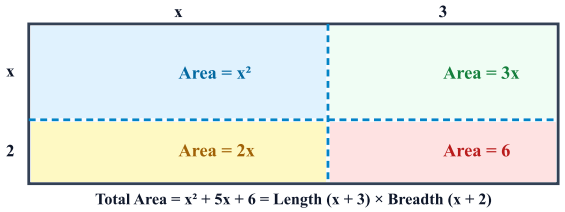
Project recording paper, ruler, pencil, eraser, pen, and Class 8 Mathematics textbook.

5. PROCEDURE

1. **Step 1 (Common Factor Method):** Express terms in irreducible factor form and factor out the greatest common divisor.
2. **Step 2 (Regrouping Method):** Rearrange four terms into pairs having common binomial factors.
3. **Step 3 (Identity Method):** Apply identities such as $a^2 - b^2 = (a - b)(a + b)$ and $(a \pm b)^2$.
4. **Step 4 (Splitting the Middle Term):** For $x^2 + bx + c$, find two numbers whose sum is b and product is c.
5. **Step 5 (Geometric Verification):** Represent $x^2 + 5x + 6$ as the area of a large rectangle of dimensions $(x + 3)$ by $(x + 2)$.

6. OBSERVATIONS AND TABULATION

S.No.	Algebraic Expression	Method Applied	Factorised Form
1	$6xy + 9y$	Common Factor (3y)	$3y(2x + 3)$
2	$ax + bx + ay + by$	Regrouping Terms	$(x + y)(a + b)$
3	$x^2 - 25$	Difference of Squares ($a^2 - b^2$)	$(x - 5)(x + 5)$
4	$p^2 + 6p + 9$	Perfect Square Identity $(a + b)^2$	$(p + 3)^2$
5	$x^2 + 5x + 6$	Splitting Middle Term ($2 + 3 = 5$, $2 \times 3 = 6$)	$(x + 2)(x + 3)$



- **Common Factor:** Reduces terms into standard products of monovariates and polynomials.
- **Geometric Model:** Proves that factorisation is finding the dimensions of a rectangle whose area is given by the trinomial.

7. CONCLUSION

- Factorisation decomposes algebraic polynomials into products of their simplest irreducible factors.
- Geometric representations confirm that algebraic identities and splitting methods correspond to physical dimensions.

8. ACKNOWLEDGEMENT

I express my sincere thanks to my **Mathematics Teacher** for continuous support and guidance during this project.

9. RESOURCES

Telangana SCERT Class 8 Mathematics Textbook (Chapter: *Factorisation*) and classroom notes.

Signature of Student
Date: ____/____/2026

Signature of Mathematics Teacher
Verified